Alibaba Qwen B2C Dataset

Anonymous production dataset:
https://github.com/alibaba-edu/qwen-bailian-usagetraces-anon

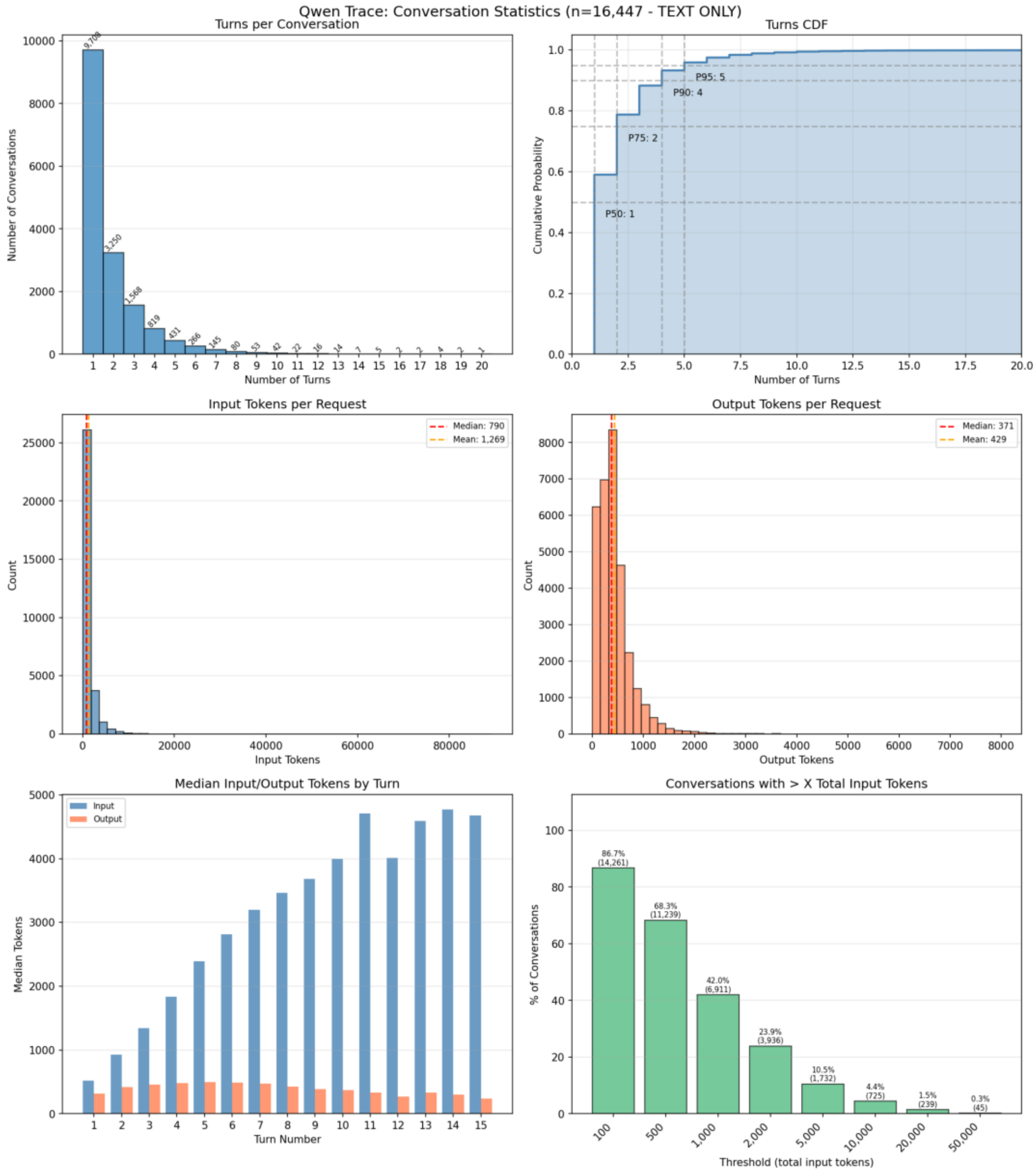
Paper: https://arxiv.org/abs/2506.02634

While this is a legitimate production trace, it is still quite old. Below we perform a comprehensive analysis of the dataset to build a better mental model of real-world multi-turn traffic distribution.

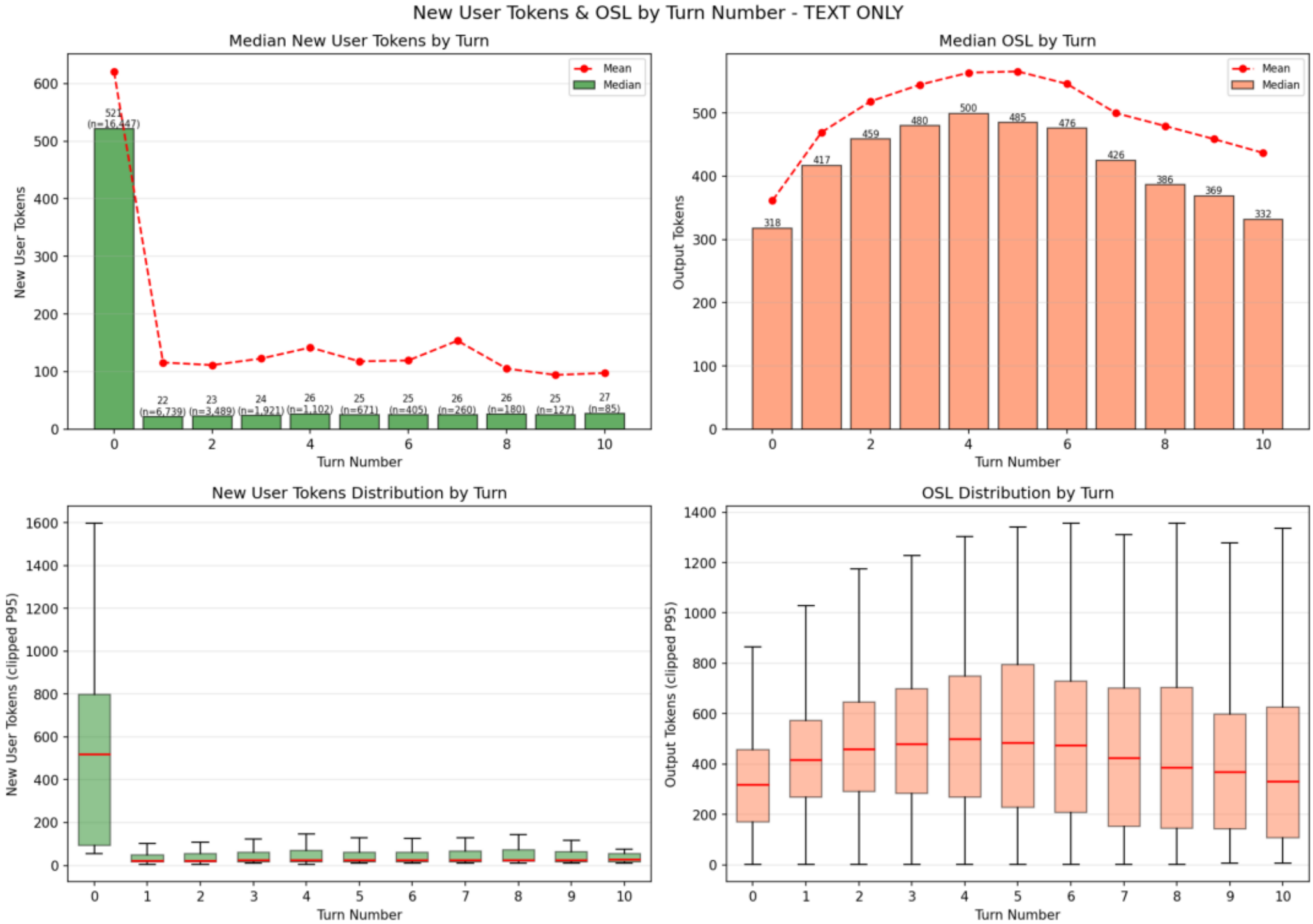
The following dataset includes multiple data formats (text, file, image). For the sake of this analysis, we will include text only.

Turn Distribution

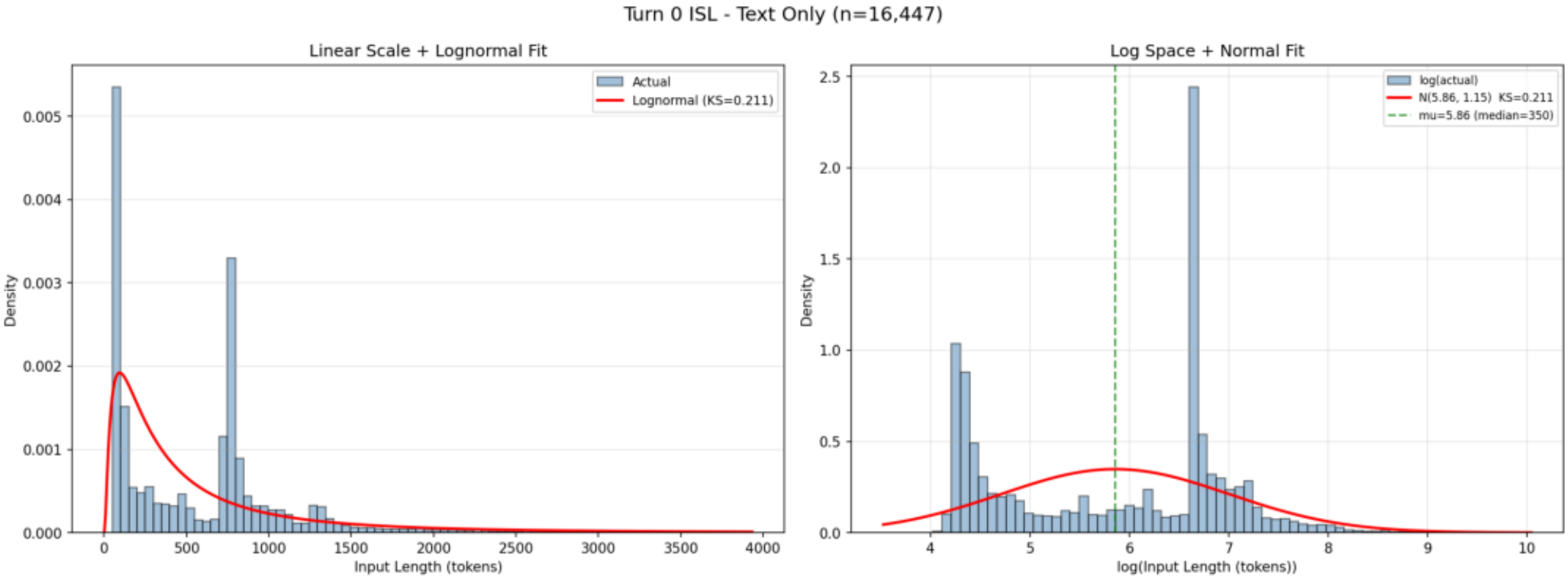
First and foremost, we see that the dataset does not have many deep conversations, with over 50% being only a single turn. We note that this is probably one of the biggest changes since this dataset was initially collected a couple years ago. Now, people are much more likely to have longer conversations as the use cases for LLMs have expanded.

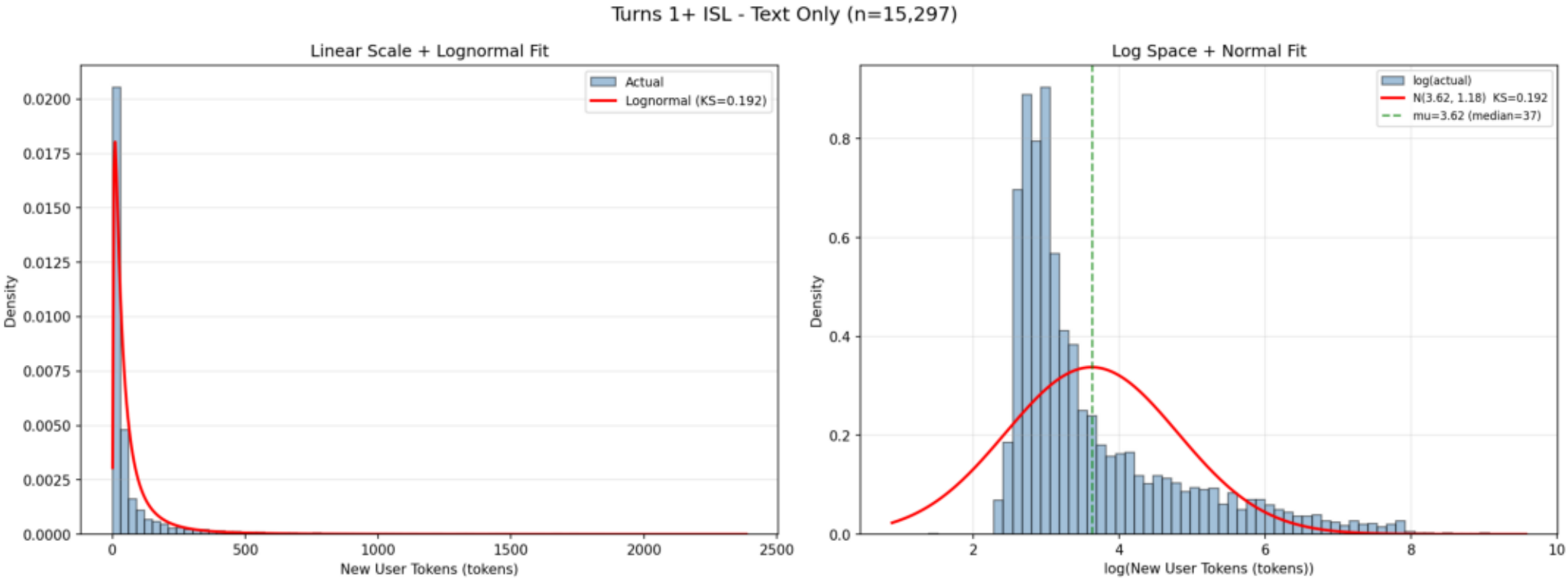

Qwen Trace: Conversation Statistics (n=16,447 - TEXT ONLY)
Turns per Conversation
Number of Conversations
Number of Turns
9,708
3,250
1,568
819
431
266
145
80
53
42
22
16
14
7
5
2
2
4
2
1
Turns CDF
Cumulative Probability
Number of Turns
P50: 1
P75: 2
P90: 4
P95: 5
Input Tokens per Request
Count
Input Tokens
Median: 790
Mean: 1,269
Output Tokens per Request
Count
Output Tokens
Median: 371
Mean: 429
Median Input/Output Tokens by Turn
Median Tokens
Turn Number
Input
Output
Conversations with > X Total Input Tokens
% of Conversations
Threshold (total input tokens)
86.7% (14,261)
68.3% (11,239)
42.0% (6,911)
23.9% (3,936)
10.5% (1,732)
4.4% (725)
1.5% (239)
0.3% (45)
100
500
1,000
2,000
5,000
10,000
20,000
50,000

The following charts describe the distribution of tokens per turn.


New User Tokens & OSL by Turn Number - TEXT ONLY
Median New User Tokens by Turn
Mean
Median
New User Tokens
Turn Number
521 (n=16,447)
22 (n=6,739)
23 (n=3,489)
24 (n=1,921)
26 (n=1,102)
25 (n=671)
25 (n=405)
26 (n=260)
26 (n=180)
25 (n=127)
27 (n=85)
Median OSL by Turn
Output Tokens
318
417
459
480
500
485
476
426
386
369
332
New User Tokens Distribution by Turn
New User Tokens (clipped P95)
OSL Distribution by Turn
Output Tokens (clipped P95)

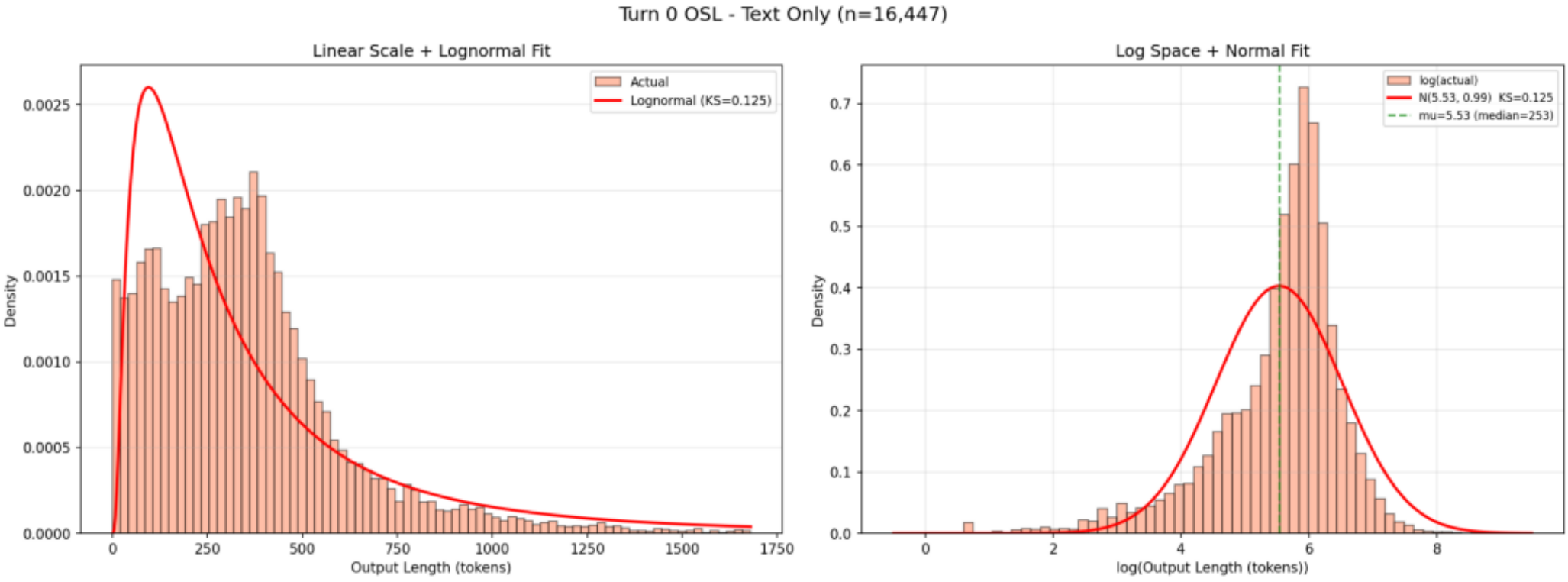
The left two charts indicate that the ISL of turn 0 is typically significantly higher than on turns 1+, but with high variance. This makes sense as users typically paste code, text, etc on the first turn and ask about it. Below is a more in depth look at the distribution of ISL for turn 0 and then turns 1+.

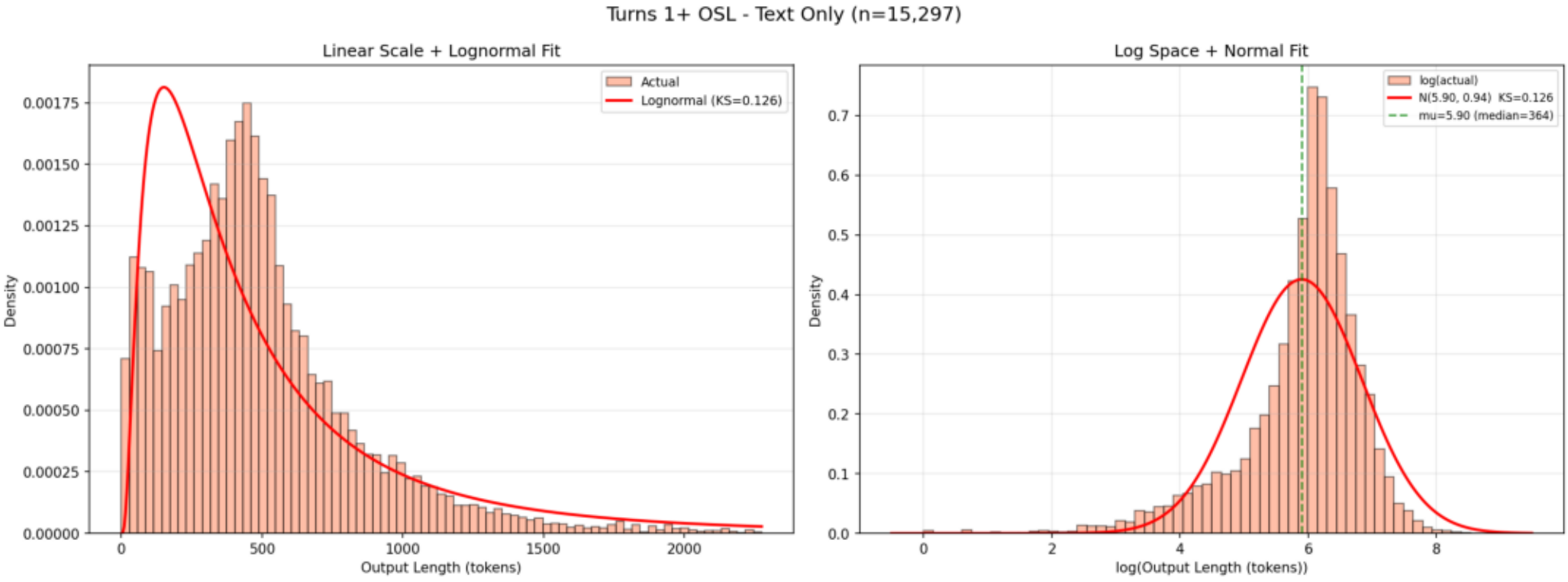

Turn 0 ISL - Text Only (n=16,447)
Linear Scale + Lognormal Fit
Actual
Lognormal (KS=0.211)
Density
Input Length (tokens)
Log Space + Normal Fit
log(actual)
N(5.86, 1.15) KS=0.211
mu=5.86 (median=350)
Density
log(Input Length (tokens))


Turns 1+ ISL - Text Only (n=15,297)
Linear Scale + Lognormal Fit
Actual
Lognormal (KS=0.192)
Density
New User Tokens (tokens)
Log Space + Normal Fit
log(actual)
N(3.62, 1.18) KS=0.192
mu=3.62 (median=37)
Density
log(New User Tokens (tokens))

Turn 0 ISL is bimodal, with some users dumping a bunch of text and querying it while others just begin with a more QA style question like "What is the capital of France?" The rest of the turns are well approximated by a lognormal distribution.

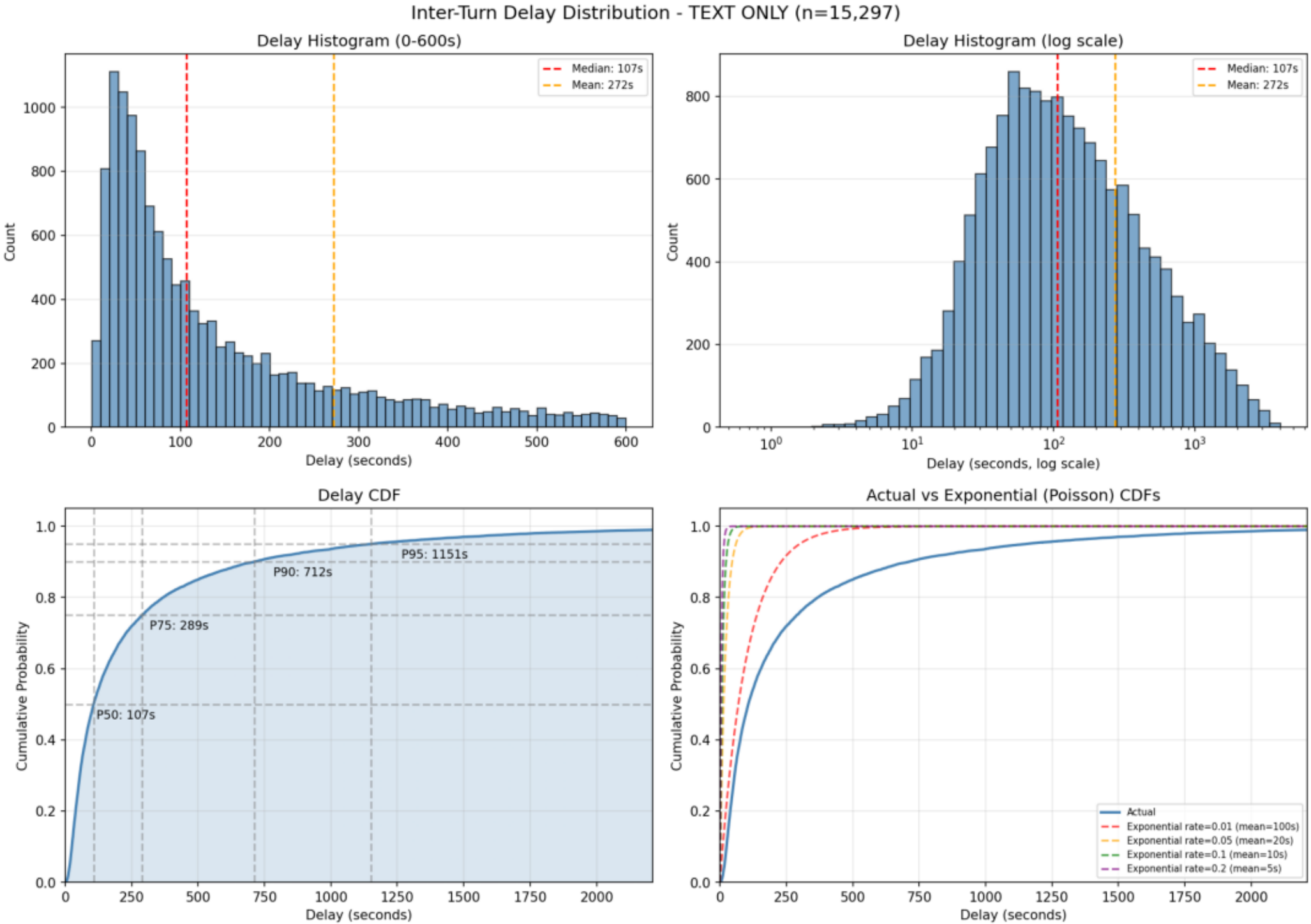
For all turns, OSL is quite well approximated by a lognormal distribution.


Turn 0 OSL - Text Only (n=16,447)
Linear Scale + Lognormal Fit
Actual
Lognormal (KS=0.125)
Density
Output Length (tokens)
Log Space + Normal Fit
log(actual)
N(5.53, 0.99) KS=0.125
mu=5.53 (median=253)
Density
log(Output Length (tokens))


Turns 1+ OSL - Text Only (n=15,297)
Linear Scale + Lognormal Fit
Actual
Lognormal (KS=0.126)
Density
Output Length (tokens)
Log Space + Normal Fit
log(actual)
N(5.90, 0.94) KS=0.126
mu=5.90 (median=364)
Density
log(Output Length (tokens))

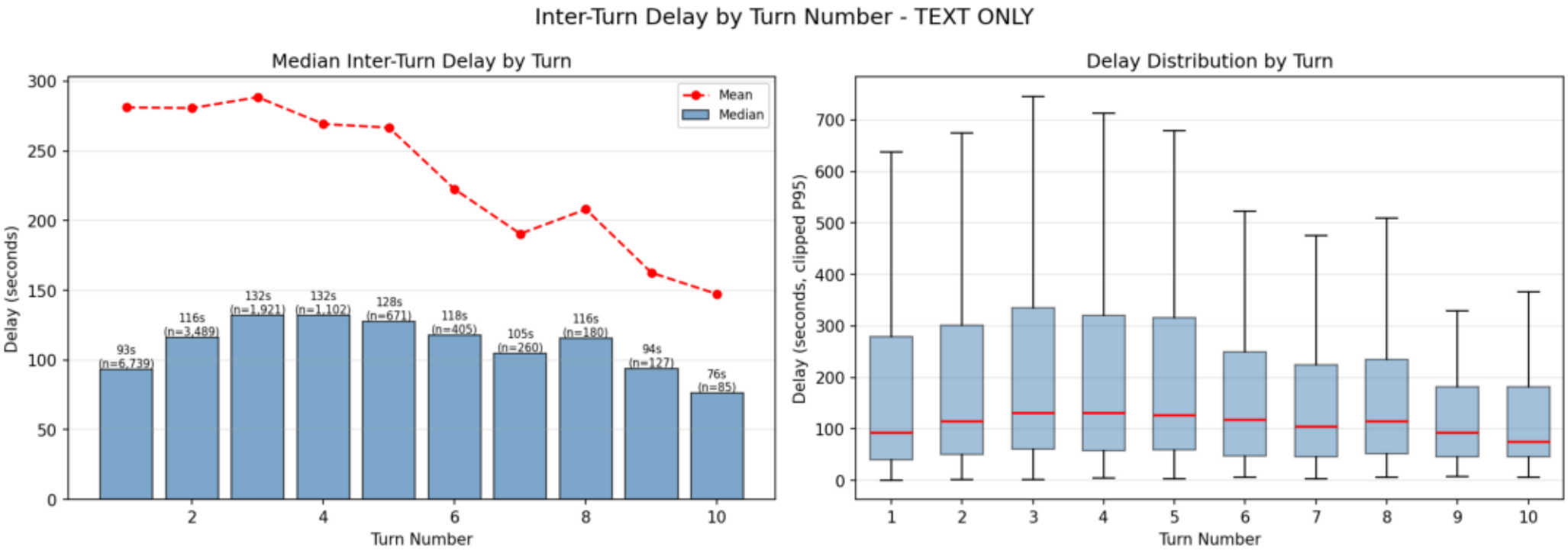
Think Time (Delay Between Turns)

What is mainly interesting is the request timing analysis we see from this dataset. Note that we are analyzing the B2C (multi-turn) dataset.


Inter-Turn Delay Distribution - TEXT ONLY (n=15,297)
Delay Histogram (0-600s)
Median: 107s
Mean: 272s
Count
Delay (seconds)
Delay Histogram (log scale)
Median: 107s
Mean: 272s
Count
Delay (seconds, log scale)
Delay CDF
P95: 1151s
P90: 712s
P75: 289s
P50: 107s
Cumulative Probability
Delay (seconds)
Actual vs Exponential (Poisson) CDFs
Actual
Exponential rate=0.01 (mean=100s)
Exponential rate=0.05 (mean=20s)
Exponential rate=0.1 (mean=10s)
Exponential rate=0.2 (mean=5s)
Cumulative Probability
Delay (seconds)

This indicates that the requests follow an approximate log-normal distribution.

We also note that median think time tends to be somewhat independent of turn number, with a slight downwards trend as conversations progress.


Inter-Turn Delay by Turn Number - TEXT ONLY
Median Inter-Turn Delay by Turn
Mean
Median
Delay (seconds)
Turn Number
93s (n=6,739)
116s (n=3,489)
132s (n=1,921)
132s (n=1,102)
128s (n=671)
118s (n=405)
105s (n=260)
116s (n=180)
94s (n=127)
76s (n=85)
Delay Distribution by Turn
Delay (seconds, clipped P95)
Turn Number